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to_array from LFTS with updates

Changes since R3

- simplify returns
- mandates and expects
- highlight insertion in wording
- add feature testing macro

Changes since R2

- switch back to `{{ a[ i ] ... }}` in specification

Introduction

This paper proposes to adopt `to_array` from Library Fundamentals TS along with an rvalue-reference overload based on the feedback from the users.

```
auto lv = to_array("foo");           // array<char, 4>
auto rv = to_array<int>({ 'a', 'b' }); // array<int, 2>
```

Discussion

In C++17 we introduced deduction guide on `std::array` :

```
auto a = array{ 1, 2, 3 };
```

But it is impossible to create an `array<char, 4>` from `array{ "foo" }` because otherwise, one won't be able to explain why `array{ "foo", "bar" }` creates `array<char const*, 2>`. So one overload set, despite whether it's `std::array` deduction guide or `make_array` from LFTSv2, is not enough to serve the purpose of "creating a `std::array` from elements" and "creating a `std::array` by copying from a built-in array" at the same time. That is why^[1] we introduced `to_array` in LFTSv2:

```
auto a1 = to_array("foo");           // array<char, 4>
auto a2 = array{ "foo" };            // array<char const*, 1>
```

For the same reason, `to_array` should be adopted into the IS.

However, the original version of `to_array` only takes an lvalue reference-to-array, and users reported^[2] some issues regarding this design. The first issue is that it becomes very hackish to deduce only the array bound from a *braced-init-list*:

```
auto x = to_array<int const>({ 2, 4 }); // array<int, 2>
```

The language treats *braced-init-list* as a prvalue of an array type when deducing against reference-to-array. The code above adds `const` to coin a reference that can bind to such a prvalue, and leaves the `const` to be stripped later. Anyhow, the code isn't doing what it says.

The second issue is that it does not support move-only elements, such as `unique_ptr`, no matter how you hack.

So we should add the rvalue-reference overload to fill this hole regarding type system.

Impact on the Standard

The following table illustrates the uses and the corresponding overload resolution after adopting the proposed addition. Given

```
int a[3] = {};  
struct A { int a; double b; };
```

use	argument	overload	result
<code>to_array("meow")</code>	<code>lvalue char const[5]</code>	<code>lvalue ref</code>	<code>array<char, 5></code>
<code>to_array({1, 2})</code>	<code>prvalue int[2]</code>	<code>rvalue ref</code>	<code>array<int, 2></code>
<code>to_array<long>({1, 2})</code>	<code>prvalue long[2]</code>	<code>rvalue ref</code>	<code>array<long, 2></code>
<code>to_array(a)</code>	<code>lvalue int[3]</code>	<code>lvalue ref</code>	<code>array<int, 3></code>
<code>to_array(std::move(a))</code>	<code>xvalue int (&&)[3]</code>	<code>rvalue ref</code>	<code>array<int, 3></code>
<code>to_array<A>({{3, .1}})</code>	<code>prvalue A[1]</code>	<code>rvalue ref</code>	<code>array<A, 1></code>
<code>to_array((int[]){3, 4})</code>	C99 compound literal	see below	<code>array<int, 2></code>

The last entry is a valuable case to study. Compound literals are lvalues in the C standards, but C does not have rvalue objects of compound types anyway. Both Clang and GCC implement C99 compound literals as extensions in C++ and treat them as prvalues. Let's take a closer look at this feature,

```
(int[]){3, 4}
```

It is an array literal formed by a C-style cast notation (like `(char)`) for an array type followed by an initializer list. In the same way, we can interpret `to_array` as

```
to_array<int>({3, 4})
```

a `std::array` literal formed by a C++ style cast notation (like `static_cast<char>(...)`) for the array element type surrounding an initializer list.

An interesting consequence of this proposal is that we no longer need `make_array`:

LFTSv2	C++20
<code>make_array(1, 2, 3)</code>	<code>array{ 1, 2, 3 }</code>
<code>make_array<long>(1, 2, 3)</code>	<code>to_array<long>({ 1, 2, 3 })</code>

In the future, we may have `array<long>{ 1, 2, 3 }` to replace this specific use of `to_array`, but that will not defeat the primary purpose of `to_array` – a facility to create `std::array` by copying or moving from a built-in array.

Supplying the bound with `to_array<T, N>({ ... })` is more strict compared to `array<T, N>{ ... }`, because brace elision in aggregate initialization can sometimes give unexpected outcomes:

```
using E = complex<double>;

// Nicol's quiz time
auto a = array<E, 10>{{1, 2}, {3, 4}};
auto b = array<E, 10>{{{1, 2}, {3, 4}}};
auto c = array<E, 10>{{1, 2}};
```

Implementation

Godbolted: <https://godbolt.org/g/GqbygN> (<https://godbolt.org/g/GqbygN>)

Wording

This wording is relative to N4820.

New section 22.3.7.6 [array.creation] (between [array.zero] and [array.tuple], which was 22.3.7.6):

22.3.7.6 Array creation functions [array.creation]

```
template<class T, size_t N>
constexpr array<remove_cv_t<T>, N> to_array(T (&a)[N]);
```

Mandates: `is_array_v<T>` is false and `is_constructible_v<T, T&>` is true .

Expects: `T` meets the *Cpp17CopyConstructible* requirements.

Returns: `{{ a[0], ..., a[N - 1] }}` .

```
template<class T, size_t N>
constexpr array<remove_cv_t<T>, N> to_array(T (&&a)[N]);
```

Mandates: `is_array_v<T>` is false and `is_move_constructible_v<T>` is true .

Expects: `T` meets the *Cpp17MoveConstructible* requirements.

Returns: `{{ std::move(a[0]), ..., std::move(a[N - 1]) }}` .

Add those signatures to 22.3.2 [array.syn]:

```
namespace std {
    // 22.3.7, class template array
    template<class T, size_t N> struct array;

    ...

    template<class T, size_t N>
        void swap(array<T, N>& x, array<T, N>& y) noexcept(noexcept(x.swap(y)));

    // 22.3.7.6, Array creation functions
    template<class T, size_t N>
        constexpr array<remove_cv_t<T>, N> to_array(T (&a)[N]);
    template<class T, size_t N>
        constexpr array<remove_cv_t<T>, N> to_array(T (&&a)[N]);
    ...
}
```

Add the following entry to Table 36 [tab:support.ft]:

Macro name	Value	Header(s)
<code>__cpp_lib_to_array</code>	<code>xxxxxxL</code>	<code><array></code>

Acknowledgements

Thank Peter Sommerlad for the inputs on this paper.

References

1. N4391 *make_array, revision 4*. <http://www.open-std.org/JTC1/SC22/WG21/docs/papers/2015/n4391.html> (<http://www.open-std.org/JTC1/SC22/WG21/docs/papers/2015/n4391.html>) ↩
2. LWG 2814 *to_array should take rvalue reference as well*. <https://cplusplus.github.io/LWG/issue2814> (<https://cplusplus.github.io/LWG/issue2814>) ↩